

EES203: Geochemistry

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Organic geochemistry

Important questions

How do organic-rich deposits form?

In which environmental settings is organic matter predominantly deposited?

What are the primary controlling factors influencing the deposition and preservation of organic matter in sedimentary environments?

General condition for OM formation

1. High primary productivity
2. Low energy depositional environment
3. Absence of dilution by inputs of inorganic mineral matter
4. Presevation rather than degradation by detritivores and decomposers

Controls on primary production

1. Photosynthesis

- Light

- Water: provides a ready source of the hydrogen needed for aerobic photosynthesis

- Temperature – solubility of CO₂ and inhibit photosynthesis by altering enzymes and cellular components

- Salinity: relatively few organisms can tolerate large fluctuations in salinity (e.g. where fresh water meets seawater in estuaries) and hypersaline conditions.

Primary productivity

ecosystem type	area (A) (10^6 km^2)	plant biomass (Gt)	annual net primary production (P) (Gt yr^{-1})	P/A ($\text{g m}^{-2} \text{ yr}^{-1}$)
tropical rain-forest	17.0	765	37.4	2200
seasonal rain-forest	7.5	260	12.0	1600
temperate forest	12.0	385	14.9	1200
boreal forest	12.0	240	9.6	800
wood/shrub land	8.5	50	6.0	700
savannah	15.0	60	13.5	900
temperate grassland	9.0	14	5.4	600
tundra and alpine	8.0	5	1.1	140
desert scrub	18.0	13	1.6	90
swamp and marsh	2.0	30	6.0	3000
lake and stream	2.0	0.05	0.8	400
open ocean	332.0	1.0	41.5	130
upwelling zones	0.4	0.008	0.2	500
continental shelf	26.6	0.27	9.6	360
algal bed and reef	0.6	1.2	1.6	2700
estuary (open water)	1.4	1.4	2.1	1500

**Estimated primary productivity and biomass for various natural environments
(sources: Whittaker and Likens 1975; DeVooys 1979)**

Aquatic productivity

Table 3.3 Estimated aquatic primary productivity (after De Vooy 1979)

type	annual net primary production		major organisms
	(Gt C yr ⁻¹)	(% total)	
kelps	0.02	0.04	laminarians
other weeds	0.01	0.02	fucoids
angiosperms	0.49	1.1	grasses, mangroves
estuaries	0.92	2.0	algae
ocean	43.5	94.9	phytoplankton
coral reefs	0.30	0.65	coraline algae
fresh water	0.58	1.3	phytoplankton

Controls on primary production

2. Stratification of the water column

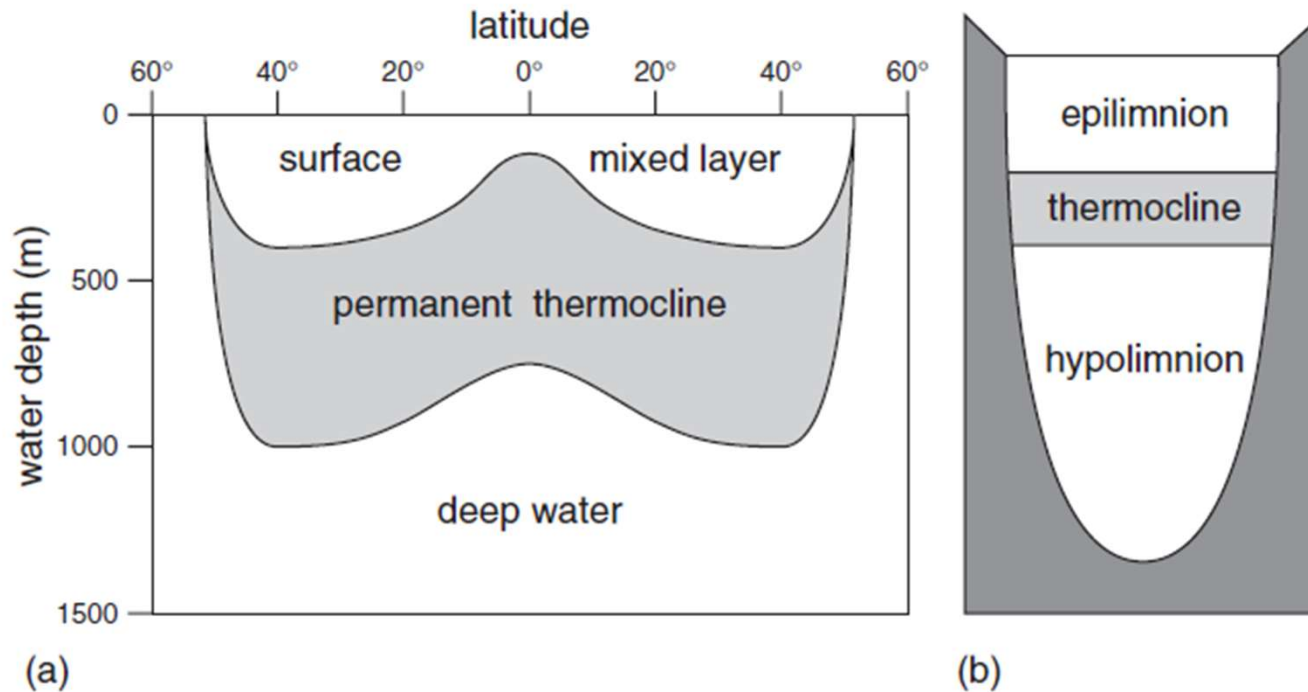
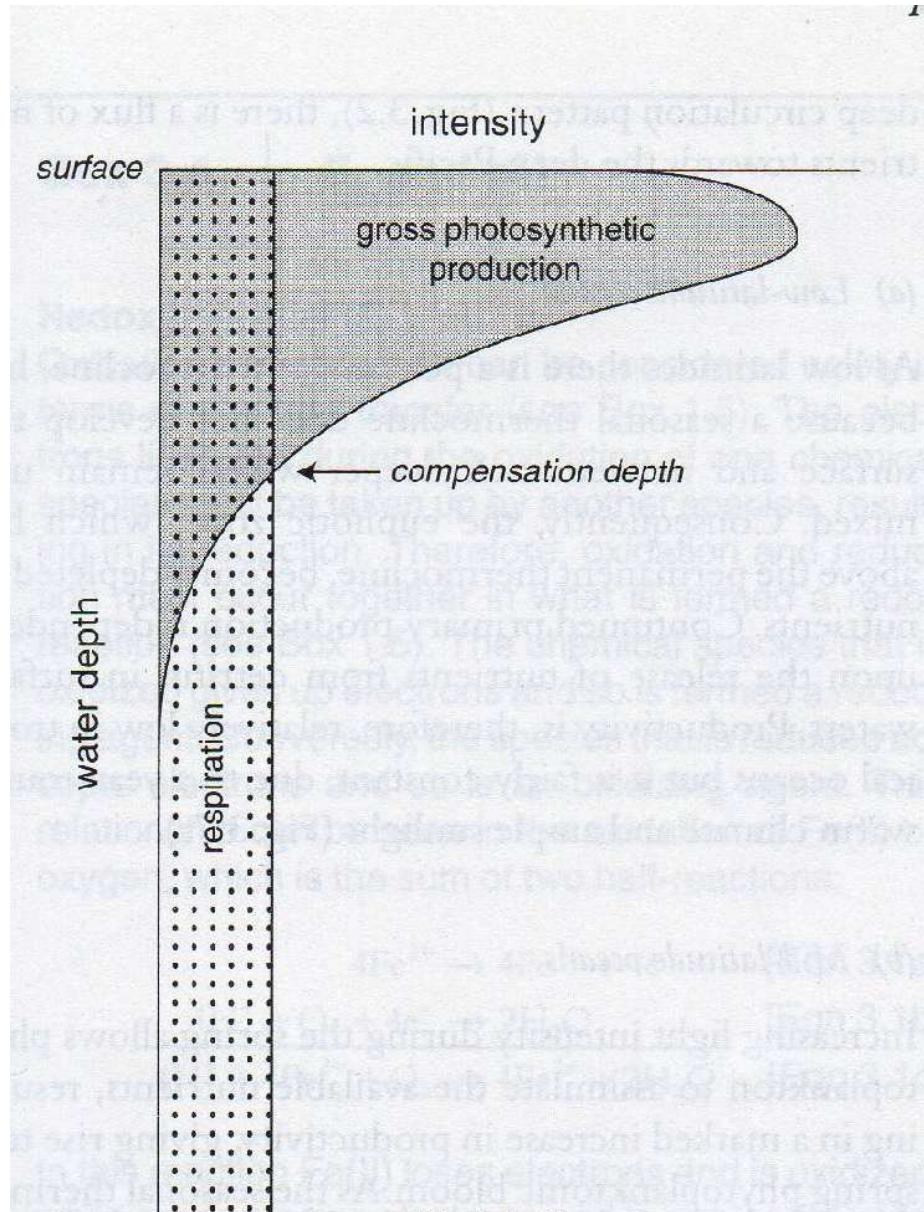


Fig. 3.1 (a) Generalized thermal stratification of the oceans; (b) corresponding nomenclature for lakes (in which the thermocline can be termed the metalimnion).

Stratification in Ocean and Lake system

Controls on primary production

3. Light



euphotic zone

Relation between phytoplanktonic production, respiration and depth (after Lalli and Parson, 1979)

Controls on primary production

4. Nutrients

category	element	ambient concentration
biolimiting elements	P, N, Si, Fe	can be almost totally depleted in surface waters
biointermediate elements	Ba, Ca, C, Ra	partially depleted in surface waters
non-limiting elements	B, Br, Cs, Cl, F, Mg, K, Rb, Na, Sr, S	show no measurable depletion in surface waters

Classification and availability of essential elements in the ocean

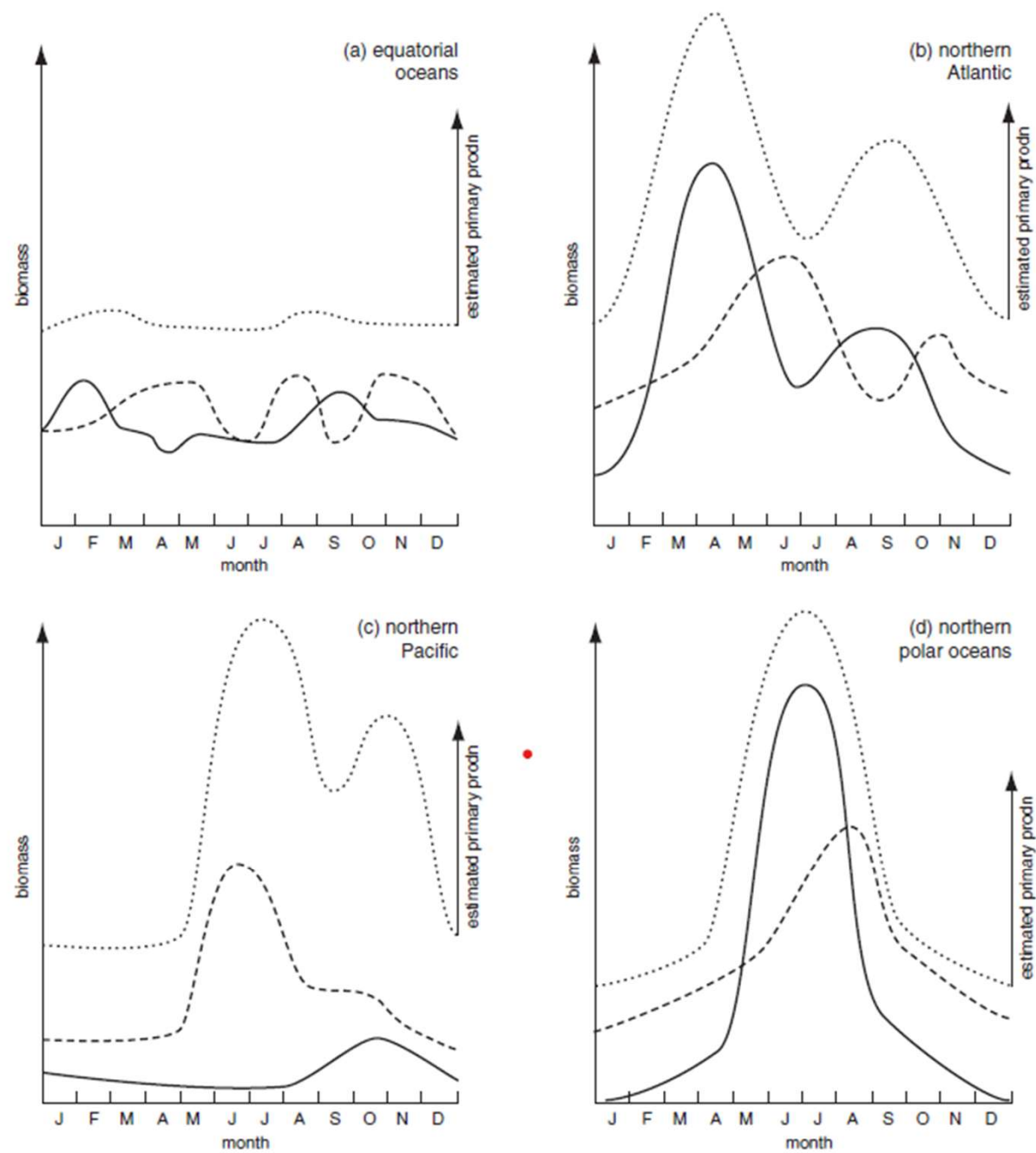


Fig. 3.7 Approximate annual standing biomass profiles for phytoplankton (solid lines) and zooplankton (broken lines) in various regions of the oceans. Estimated phytoplanktonic production is indicated by dotted lines (after Heinrich 1962; Parsons et al. 1977).

Controls on primary production

4. Nutrients

a) Low latitude ocean : Low productivity

b) Mid-latitude oceans: spring phytoplankton bloom

c) High latitude oceans: Light intensity available only during 5-6 months
single large algal bloom during the summer

d) Stratified lakes: spring phytoplankton bloom

Spatial variation in marine primary production

- The productivity is not uniform over the surface of oceans due to the effect of upwelling and downwelling on nutrient availability
- The degree of flooding of continental margins is also important in terms of the area of productive coastal waters
- Nutrient supply from the land (e.g., rivers) contribute to the productivity of the shelf areas

Spatial variation in marine primary production

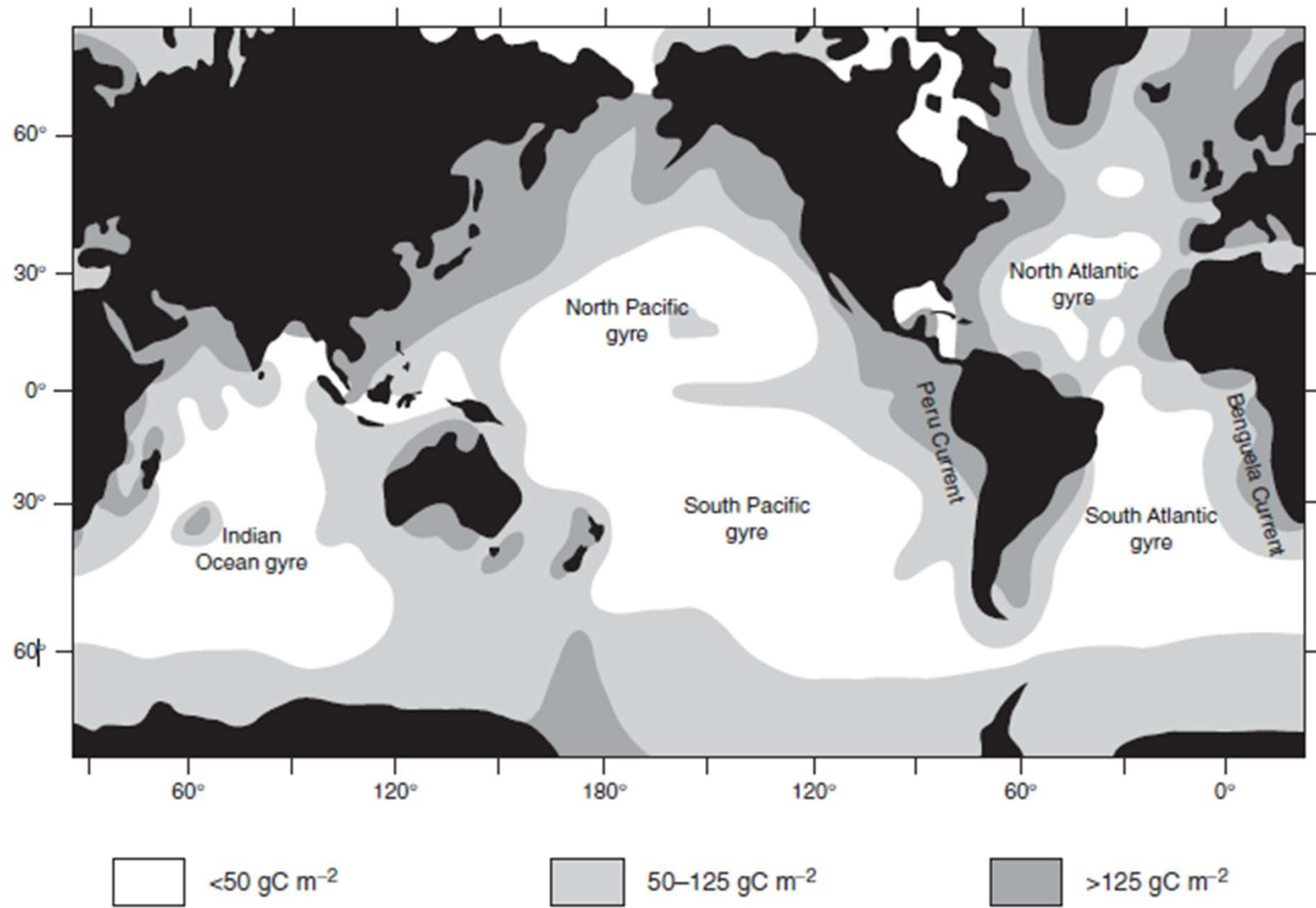


Fig. 3.8 Estimated annual production in the oceans (after several sources, including FAO 1972; Koblenz-Mishke et al. 1970).

Spatial variation in terrestrial productivity

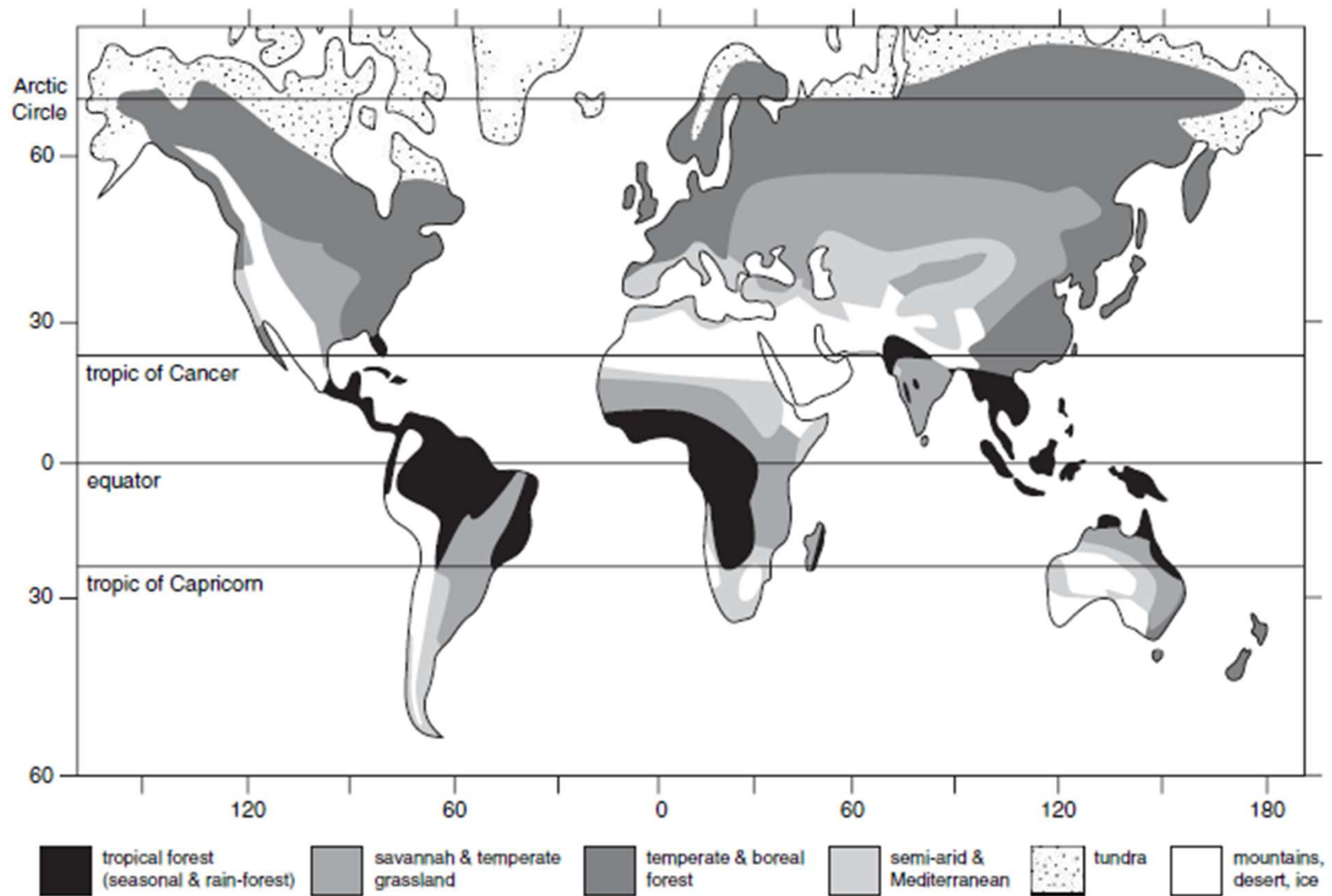
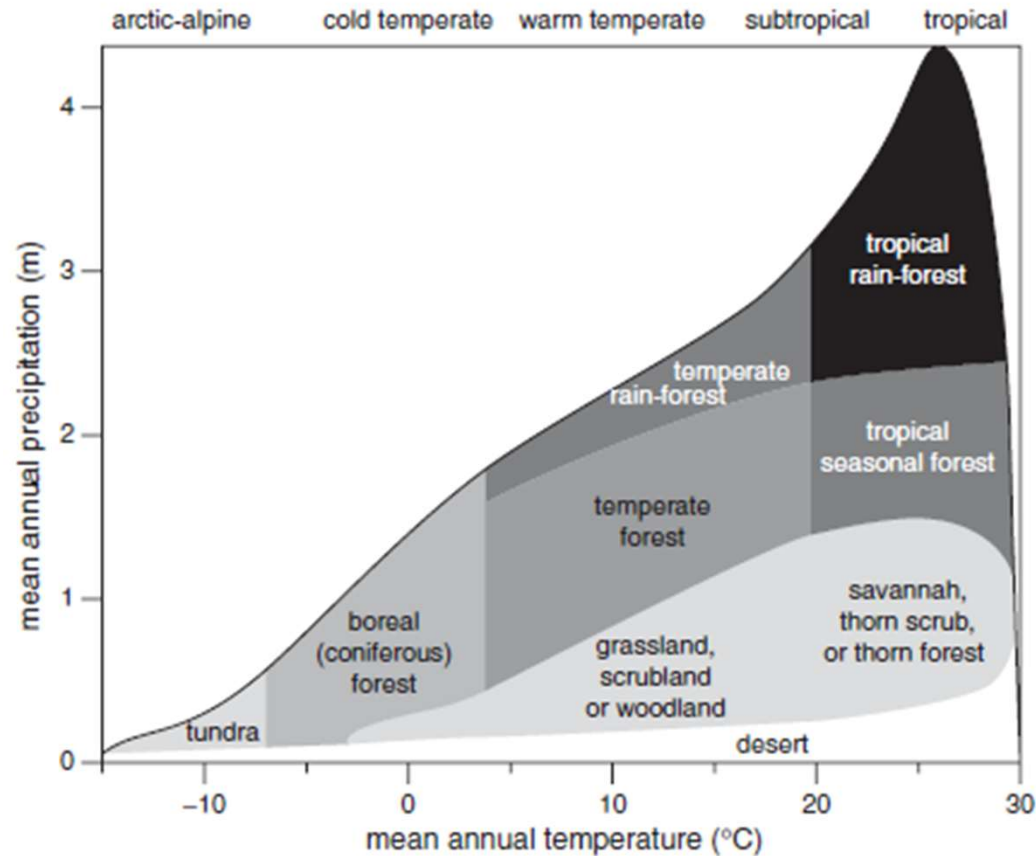


Fig. 3.13 Terrestrial biomes (after Cox & Moore 1993).

Variation in higher plant populations

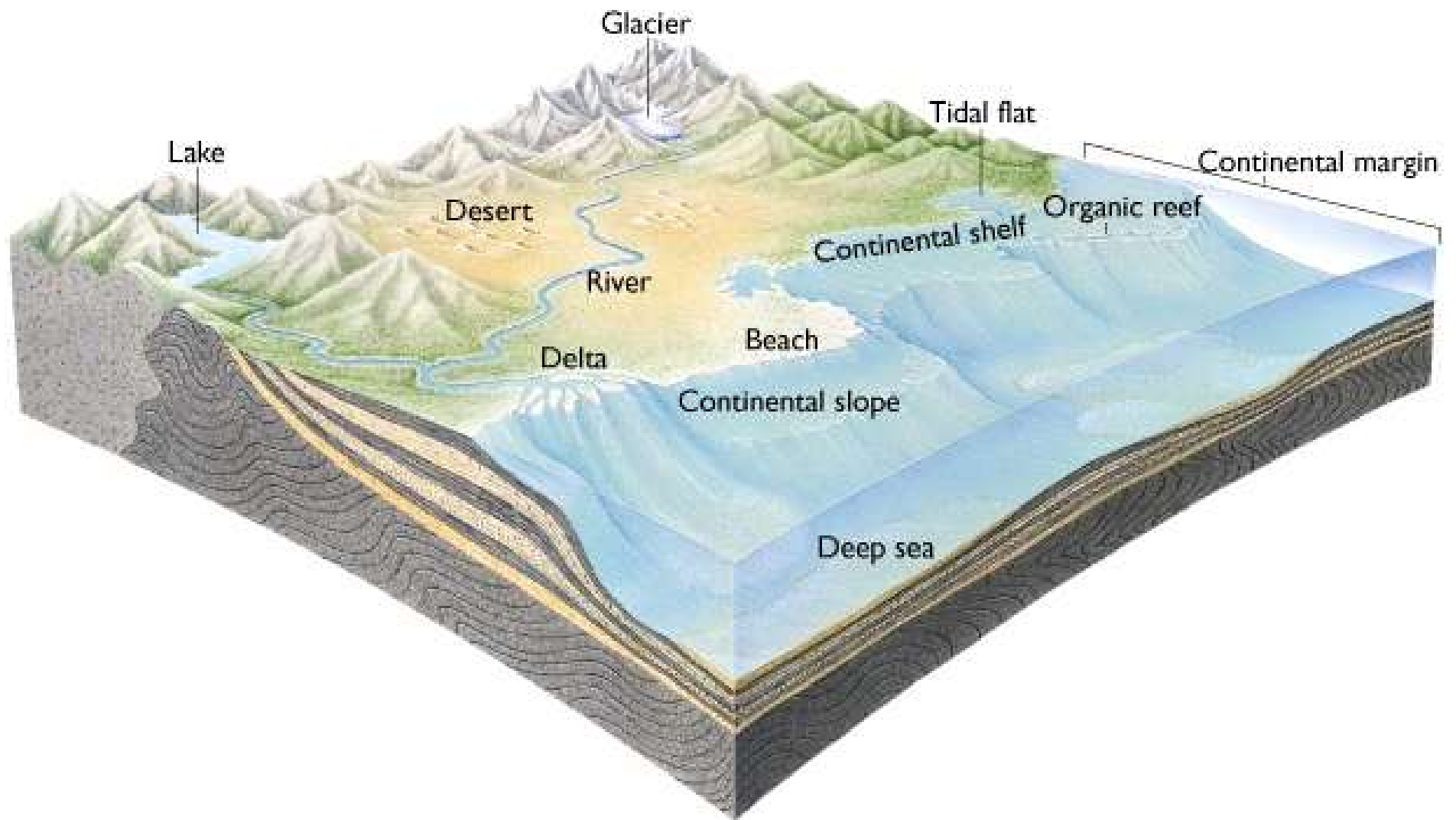


Approximate correlation of terrestrial biomes with temperature and precipitation (after Whitaker 1975)

Factors affecting sedimentary preservation of organic matter

- The flux of organic carbon to the sea floor is approximately proportional to primary production and inversely proportional to water depth
- Stratification of the water column is a prerequisite for organic preservation.

Depositional environment associated with accumulation of organic matter



Depositional environment associated with accumulation of organic matter

Table 3.8 Approximate proportions of sedimentary organic carbon accumulating in various present-day marine depositional environments (based on a total C_{org} burial rate of 160 Mt C yr^{-1} ; after Hedges & Keil 1995, based on Berner 1989)

environment	%
delta	44
shelf and upper slope	42
shallow-water carbonate	4
high-productivity slope	4
high-productivity pelagic	2
low-productivity pelagic	3
anoxic basin	<1

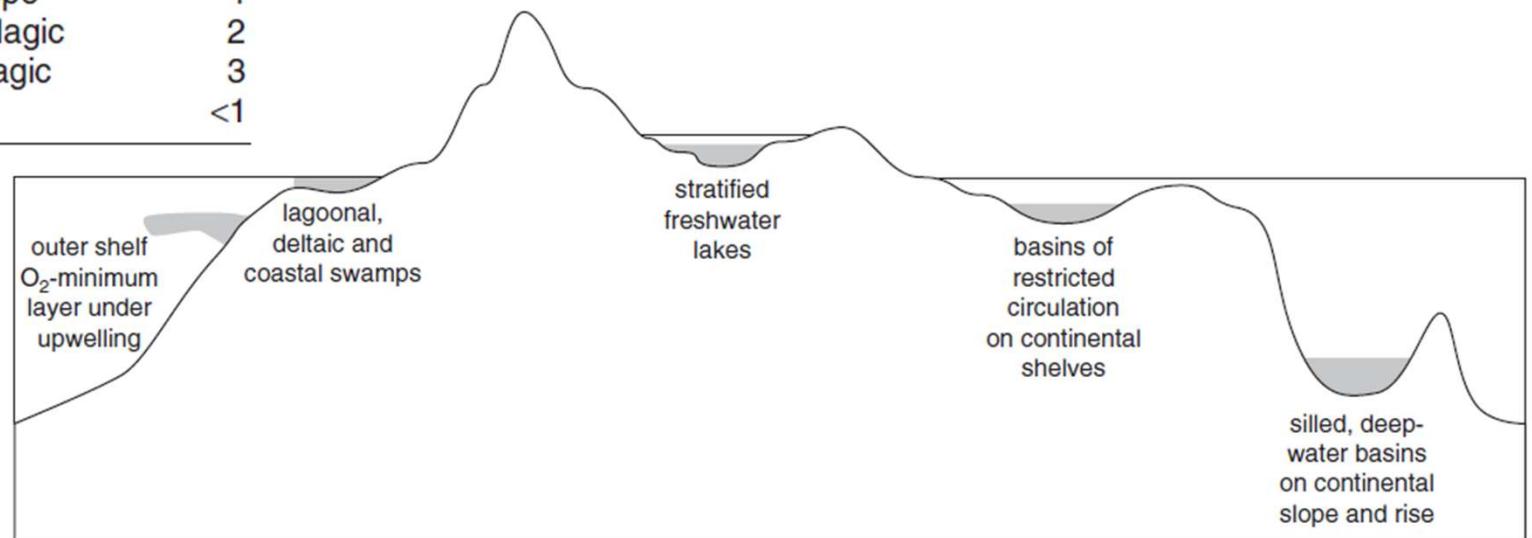
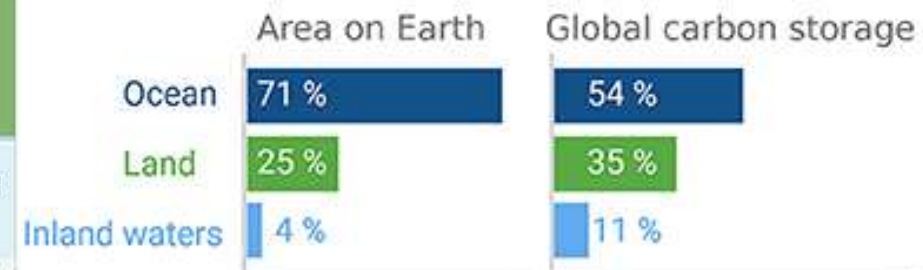
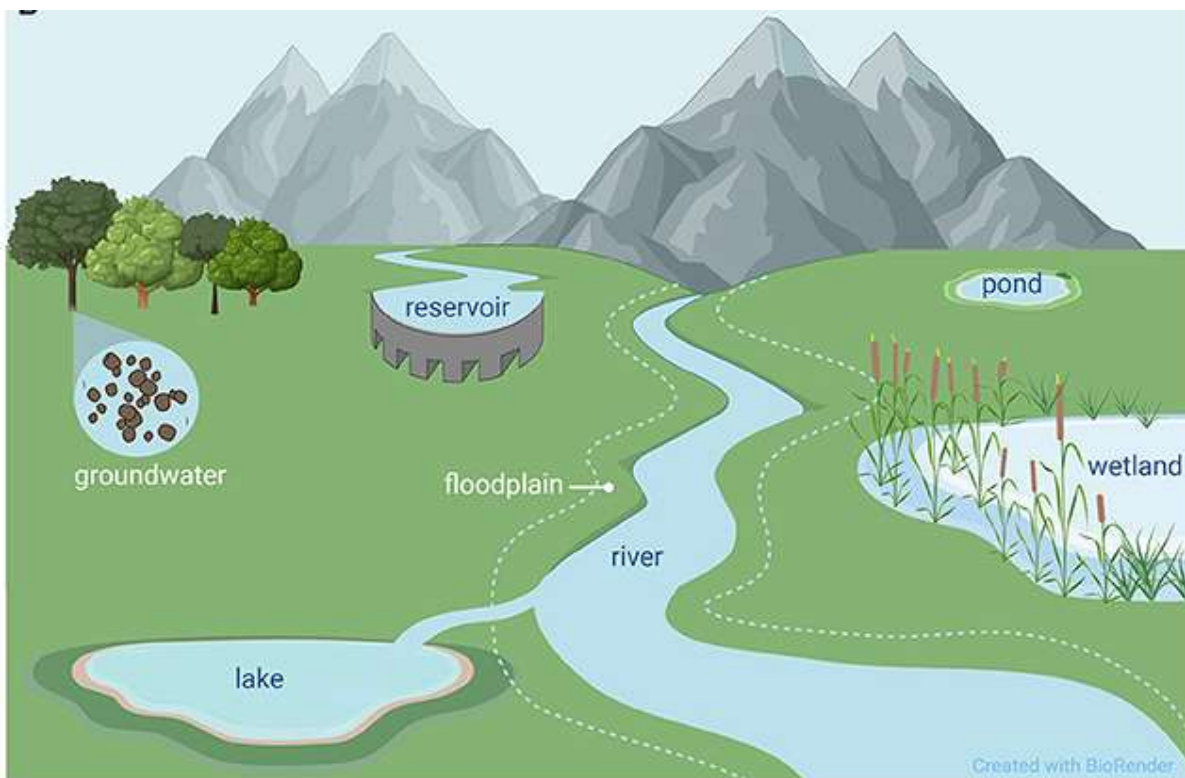


Fig. 3.23 Important oxygen-depleted environments (shaded areas) associated with deposition of organic-rich sediments (after Brooks et al. 1987).

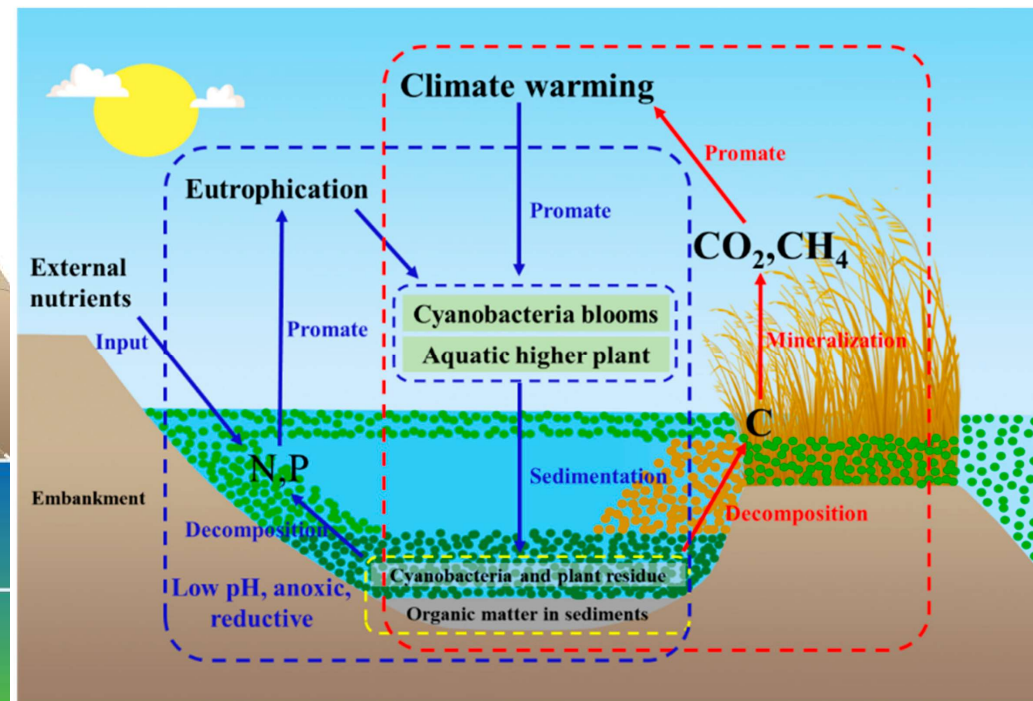
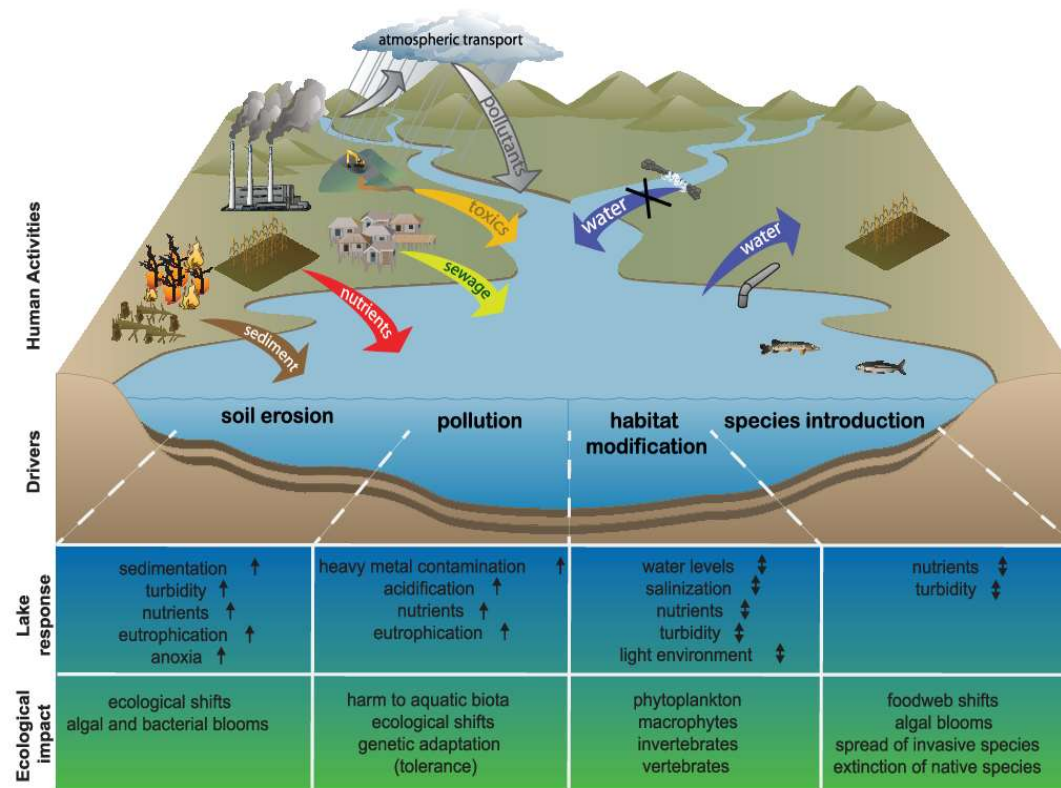
Important oxygen depleted environment associated with deposition of organic rich sediments
(after Brooks et al. 1987)

Terrestrial inland water and carbon storage



Source: Cisterna et al., 2024

OC transport and transformation pathways in lakes



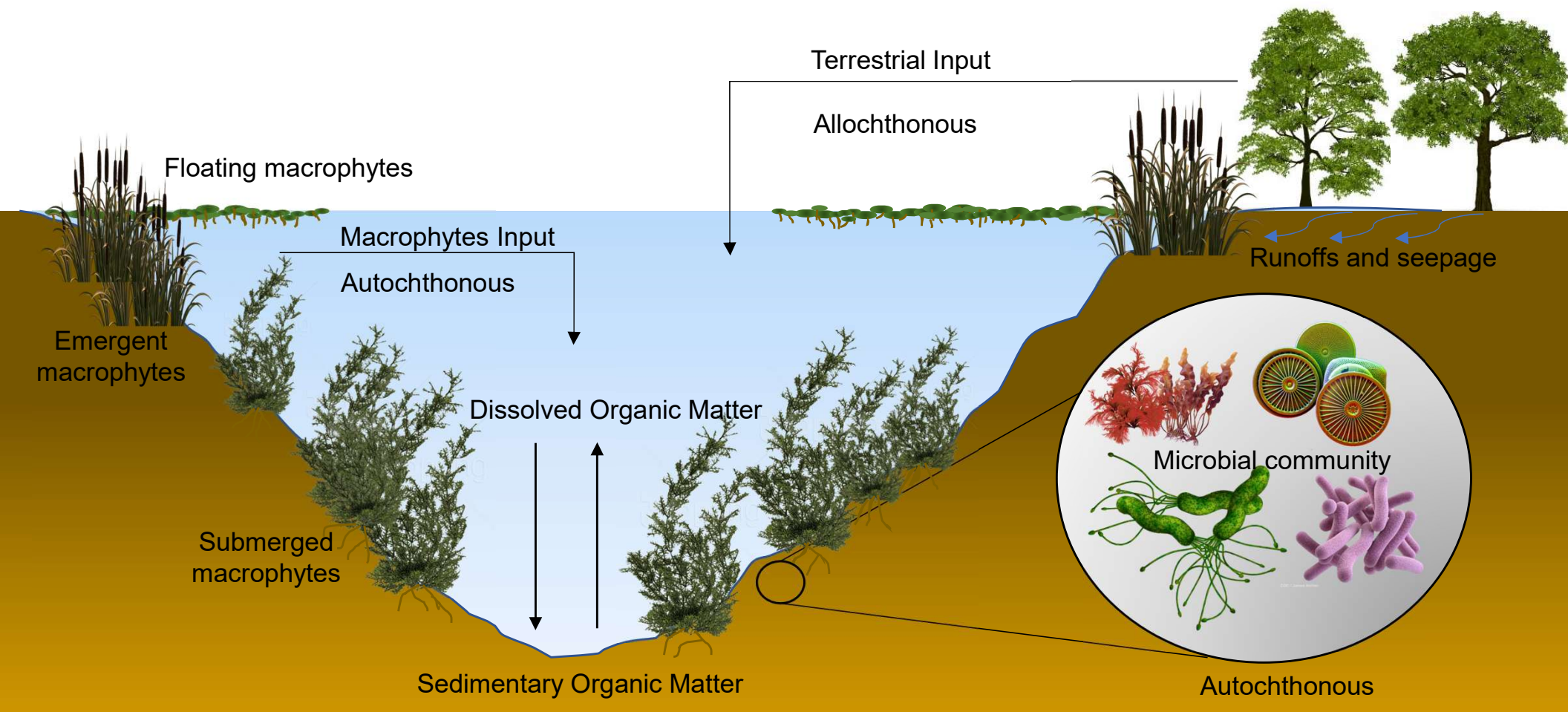
Source: Zhang et al., 2023

Depositional environment associated with accumulation of organic matter

1. Lacustrine environments

- Closed and open lakes
- Autochthonous and allochthonous OM sources

OM sources



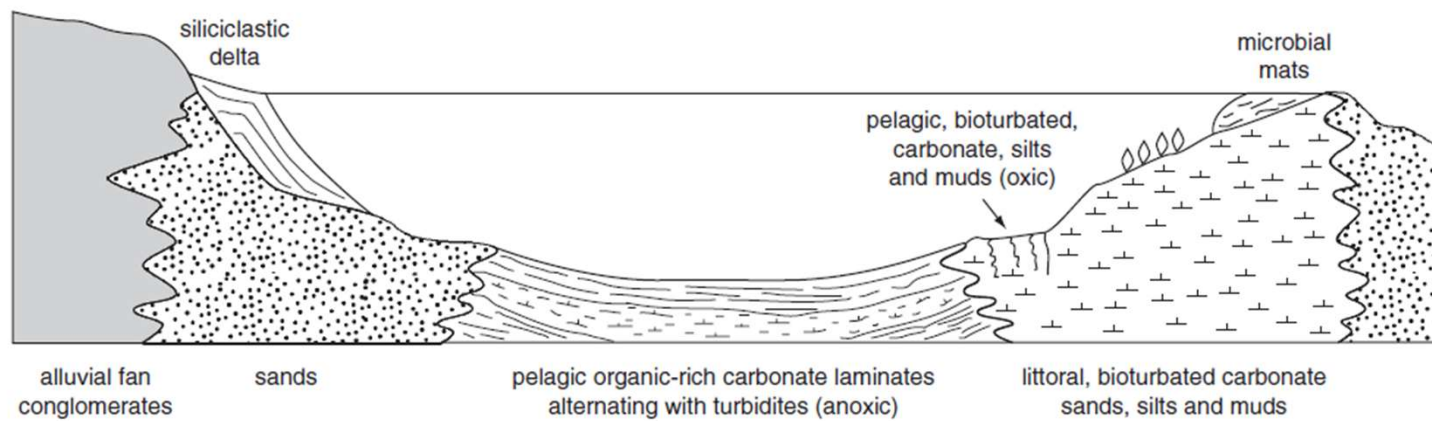


Fig. 3.27 Generalized depositional components in a hydrologically open, freshwater lake (after Eugster & Kelts 1983).

Composition of organic matter

- Aquatic organic matter from phytoplankton is rich in N due to its high protein and lipid content, and has low C/N ratios (commonly between 4 and 10).
- Terrestrial organic matter is dominated by fibrous tissues, cellulose and lignin, which are N-poor, and have C/N ratios of 20 or higher. The C/N ratio is here used to distinguish changes in aquatic and terrestrial organic matter sources in the lake (Meyers, 2003).
- In peatlands, the C/N ratio is an indicator of may indicate changes in the type of peat-forming plants (Kuhry and Vitt, 1996).

Organic matter source identification: Lacustrine and oceanic setting



Composition of C/N in modern lake system

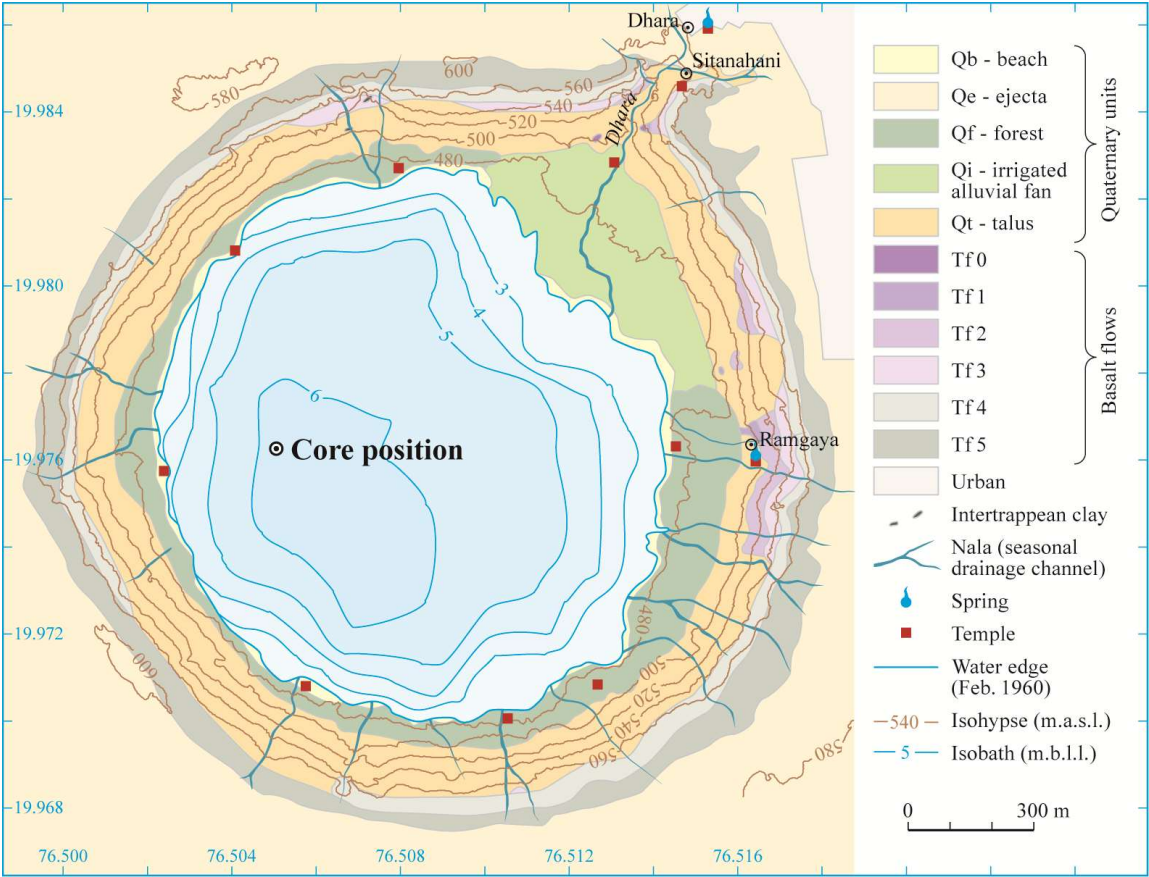
Sample label	Organic matter source	C/N
A	Macroalgae	14.88
T	Mixed plankton	10.86
H1	Herbaceous	24.68
H2	Herbaceous	25.82
H3	Herbaceous	19.49
L1	Lichen	63.55
L2	Lichen	65.73
S	Surface soil	16.99
VV	Animal excrement	21.54

Lake Meidsee (C/N content)

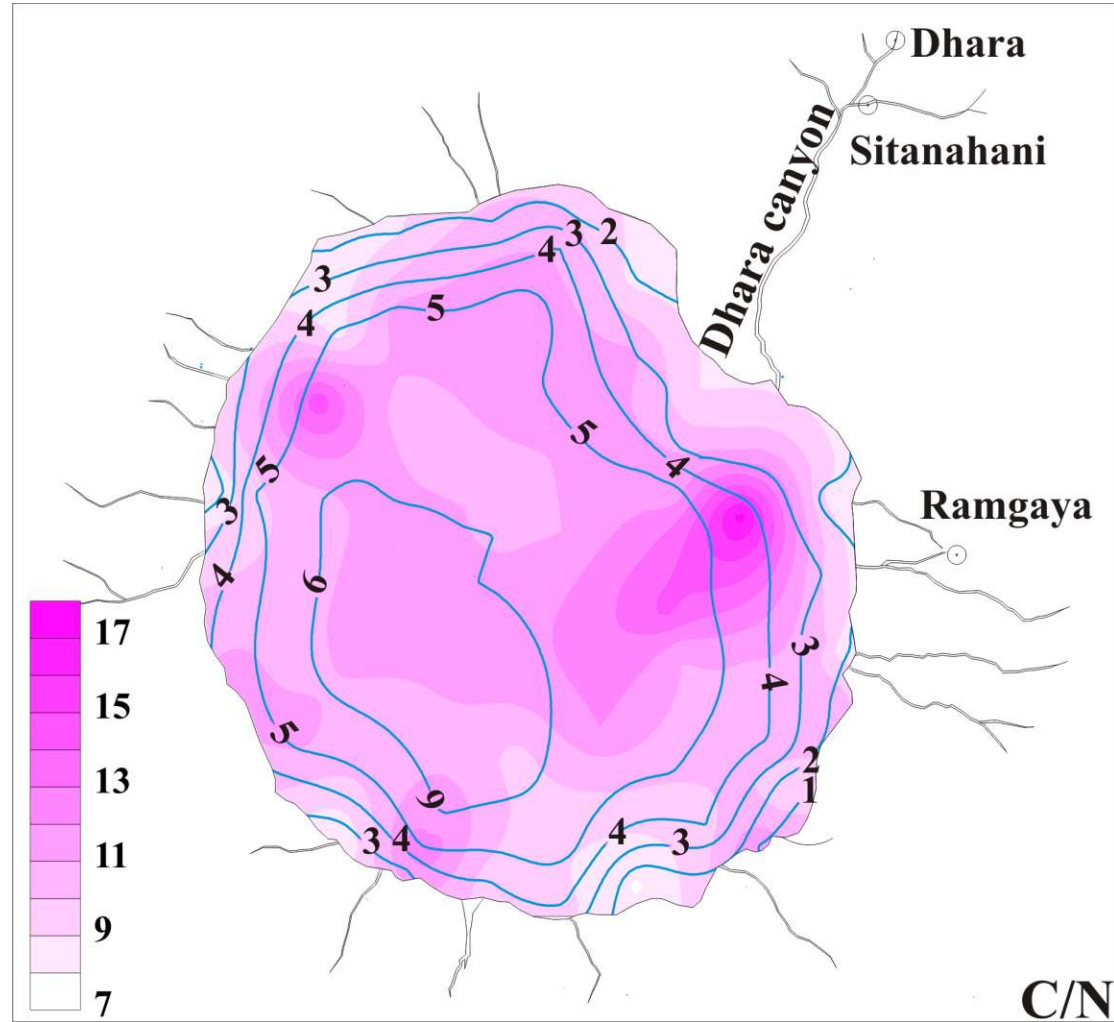
(Thevnon et al., 2012)

C/N ratios in a sediment core from Lonar lake, central India —usefulness as an organic source indicator

Lonar lake, Central India

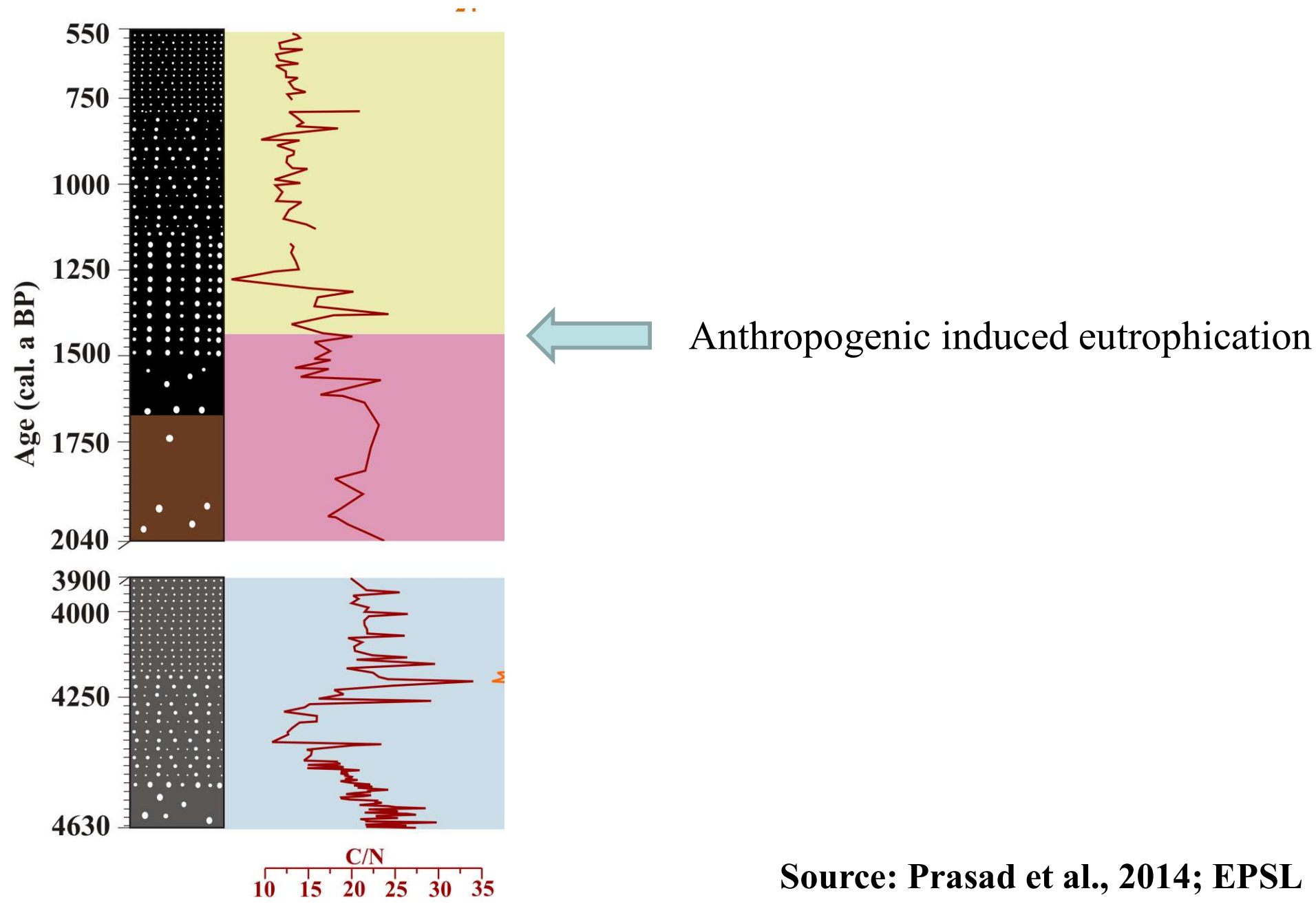


Modern C/N values in the Lonar lake



Source: Basavaiah et al., 2014; FAL

**C/N ratios in a sediment core from Lonar lake, central India —usefulness
as an organic source indicator**



Source: Prasad et al., 2014; EPSL

Depositional environment associated with accumulation of organic matter

2. Coal forming mires

Accumulation of peat in mires requires

- a) Gradual and continuous rise in groundwater table**
- b) Protection of mire against flooding by rivers and seas**
- c) Low relief hinterland limiting supply of fluvial sediments**